



Basic Principles of Medicine 1
Module: Musculoskeletal System | DLA

Introduction to Arteries, Veins and Lymphatics (DLA)

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Recommended Reading

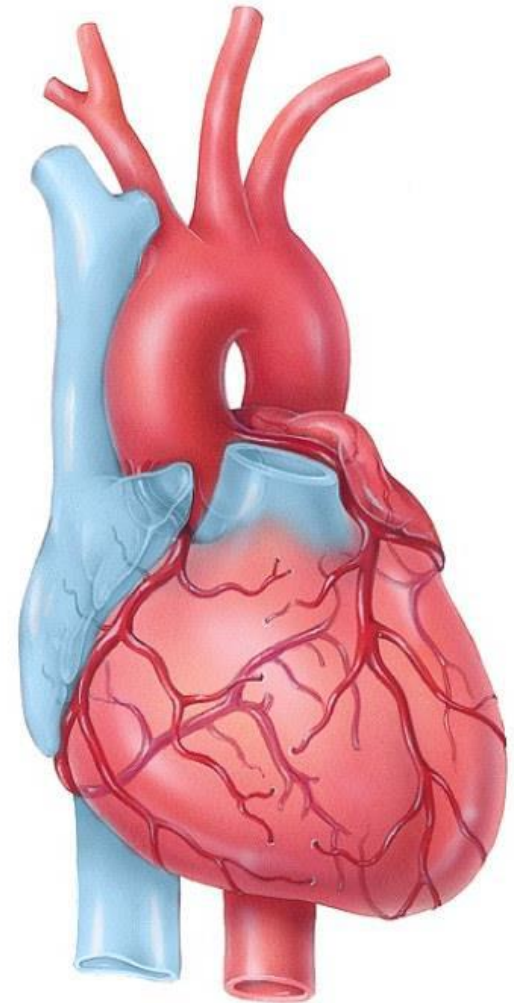
- Drake, Vogl, Mitchell. Gray's anatomy for students. 4th ed. Elsevier Churchill Livingstone.
- Printed Textbook: Chapter 1, pg. 25 - 29
- Online Textbook: Chapter 1
 - Body systems
 - Cardiovascular system
 - Lymphatic system
- Pay special attention to the green “In the Clinic” boxes for the clinical correlations for this topic

Objectives

SOM.MKII.BPM1.1.MSK.1.ANAT.1037	List the components of the cardiovascular system
SOM.MKII.BPM1.1.MSK.1.ANAT.1038	Describe the basic structure and function of arteries, veins and lymphatic vessels
SOM.MKII.BPM1.1.MSK.1.ANAT.1039	Describe the flow of blood in the cardiovascular system
SOM.MKII.BPM1.1.MSK.1.ANAT.1040	Describe the basic structure and function of lymph nodes
SOM.MKII.BPM1.1.MSK.1.ANAT.1041	Describe the flow of lymph
SOM.MKII.BPM1.1.MSK.1.ANAT.1042	Describe the terms anastomoses and collateral circulation

Cardiovascular System

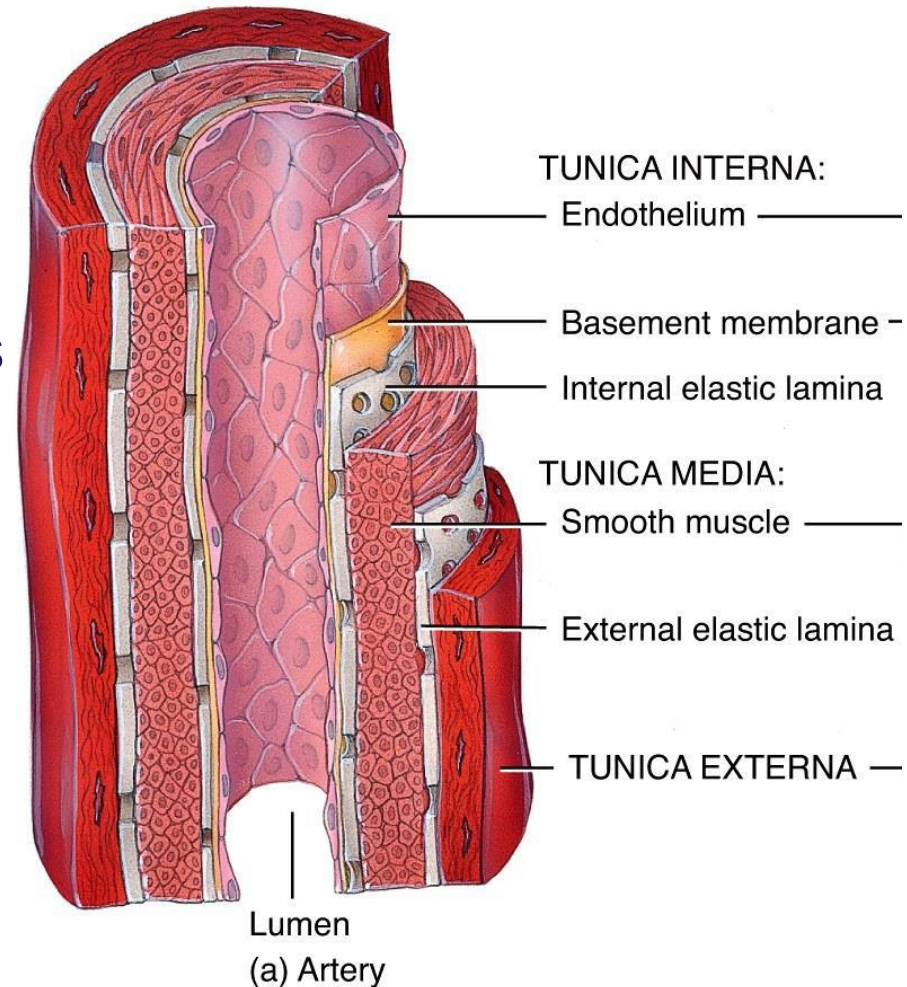
- Consists of the **heart**, which pumps blood throughout the body, and the blood vessels, which are a network of small tubes that transport the blood.
- Types of blood vessels:
 - **Arteries**, which transport blood away from the heart. Arteries have relatively thick walls that can withstand the high pressure of blood ejected from the heart.
 - **Veins**, which transport blood toward the heart. Compared to arteries, veins are thin-walled vessels with large and irregular lumens.
 - **Capillaries**, which connect the arteries and veins. It is at this level where oxygen, nutrients and waste are exchanged within the tissues.
- Walls of blood vessels
 - **Tunica intima** – inner endothelial lining
 - **Tunica media** – middle smooth muscle layer. May contain elastic fibers
 - **Tunica externa (adventitia)** – outer connective tissue layer



Cardiovascular System

Arteries

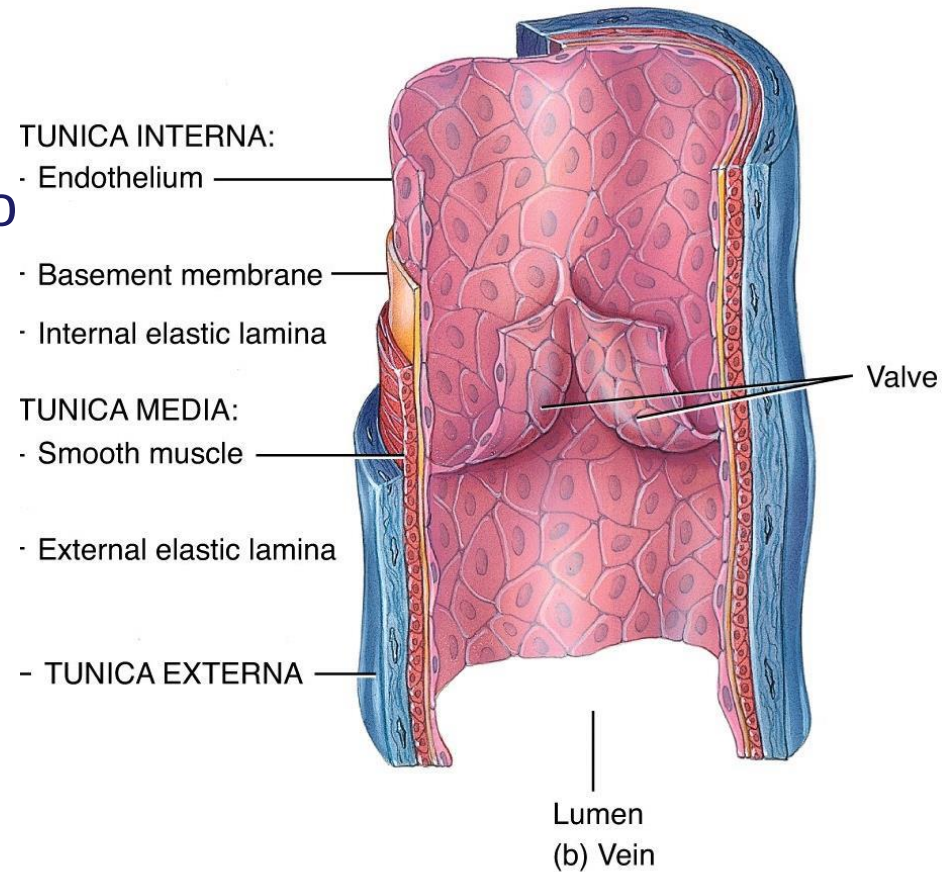
- Thick walls
 - **Tunica intima** – inner endothelial lining
 - **Tunica media** – numerous layers of smooth muscle and elastic fibers.
 - **Tunica externa (adventitia)** – Outer connective tissue layer
- Further subdivided into three classes:
 - Large
 - Medium
 - Small



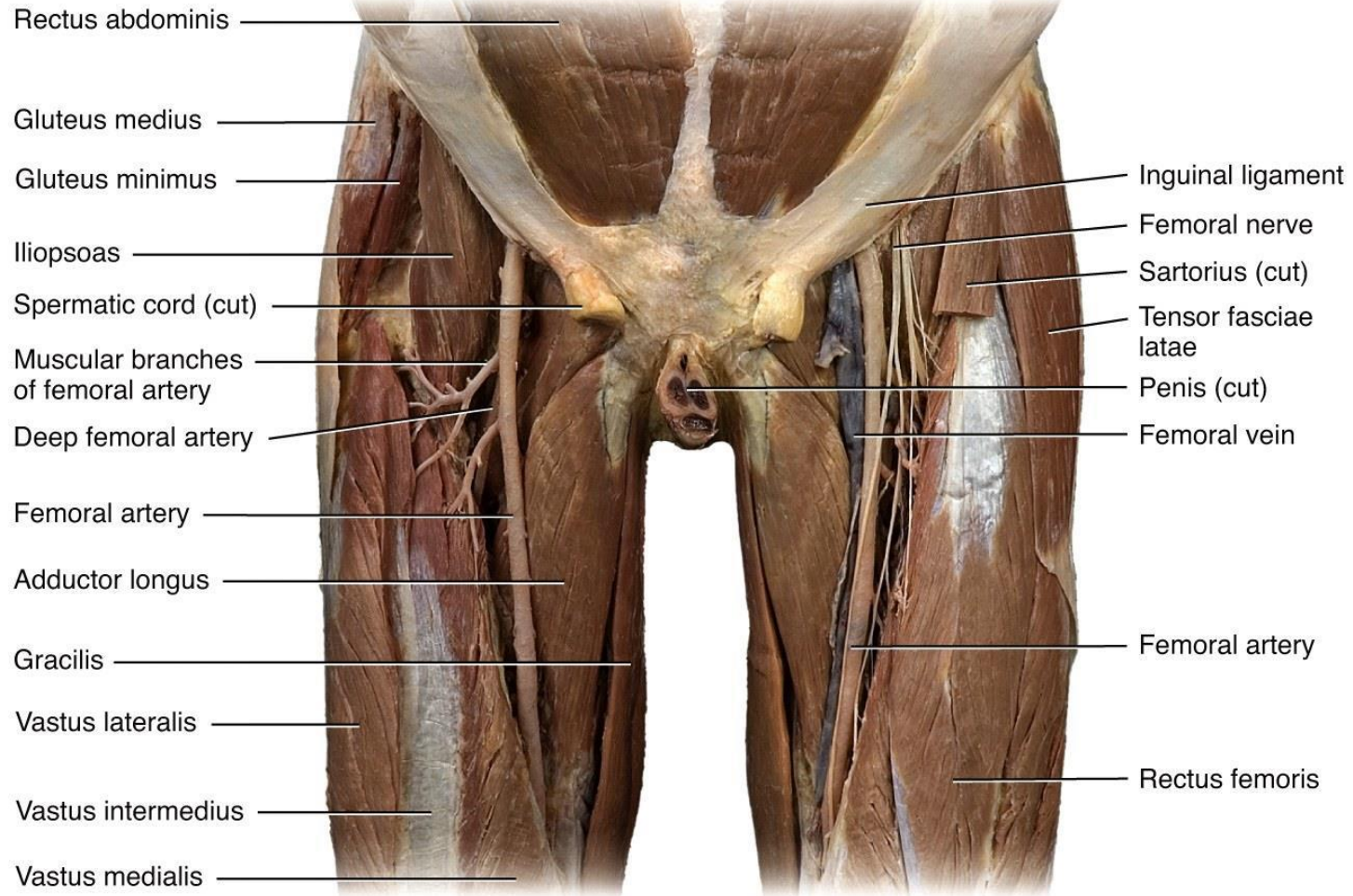
Cardiovascular System

Veins

- Thin walls
 - **Tunica intima** – inner endothelial lining, folded to form valves
 - **Tunica media** – limited layers of smooth muscle and elastic fibers.
 - **Tunica externa (adventitia)** – Thick outer connective tissue layer
- Further subdivided into three classes:
 - Large
 - Medium
 - Small



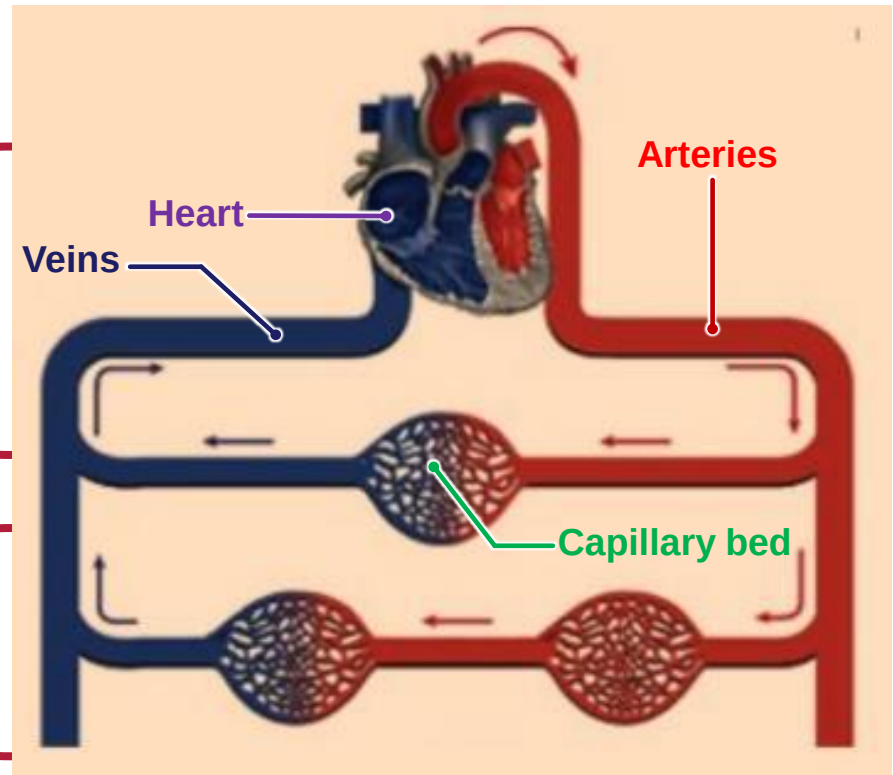
Comparison of arteries and veins in dissected cadaver



(d) Anterior view of arteries of thigh with right side deep and left side more superficial

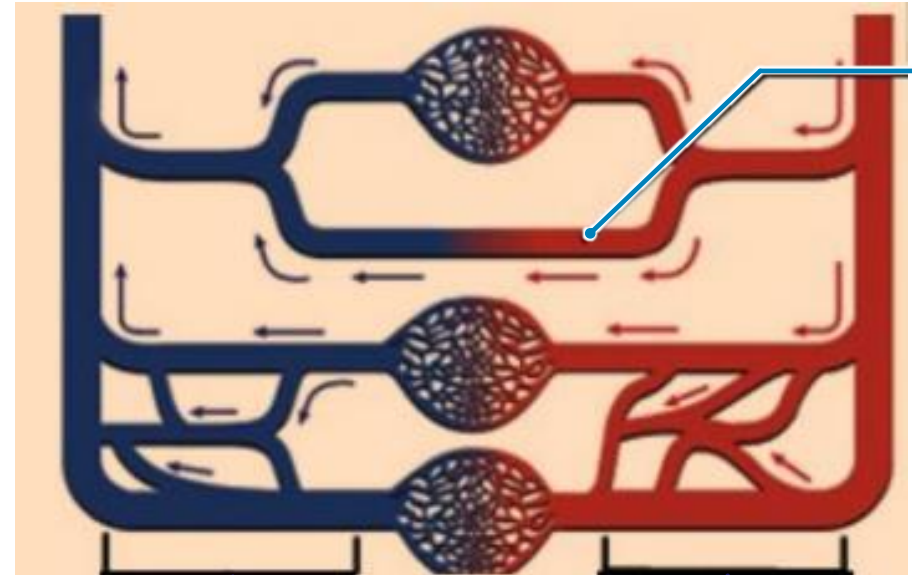
Circulatory Routes

- Most common
 - Heart – arteries – arterioles – capillary bed – venules – veins – heart
- Portal system
 - Blood flows through to consecutive capillary beds before returning to the heart
 - Between the intestines and liver
 - Anterior pituitary



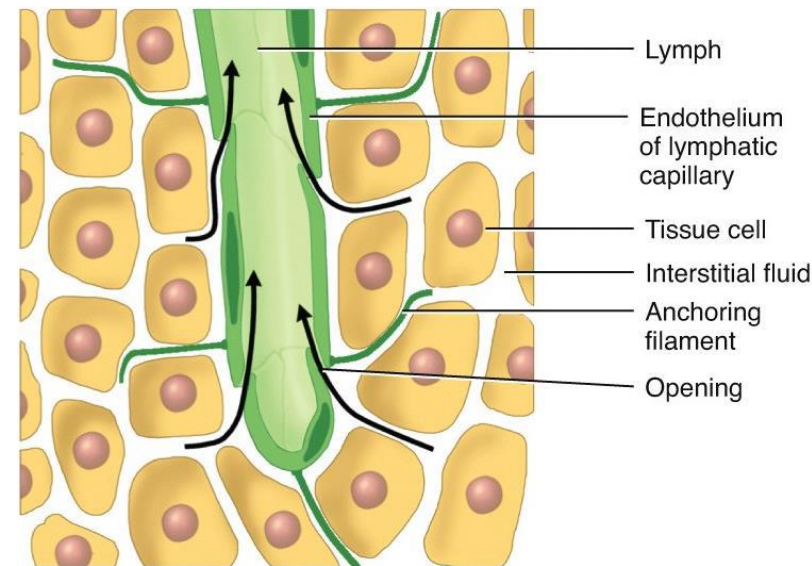
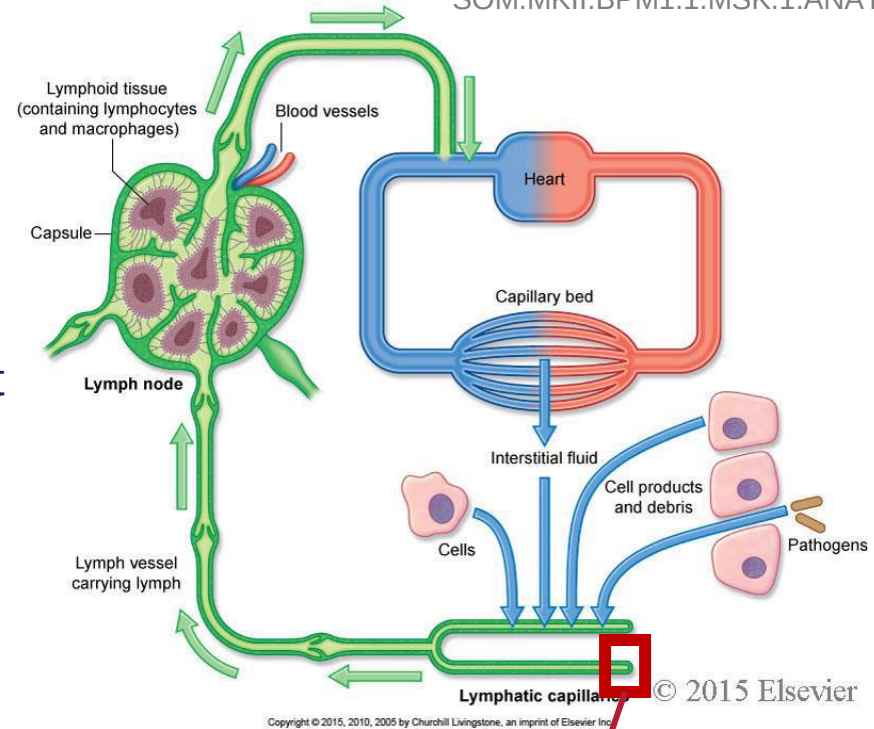
Circulatory Routes

- Arteriovenous shunt
 - Artery directly flows into a vein
 - Found in fingers, toes, ears; decreases heat loss, allows blood to bypass exposed areas during cold
- Anastomosis: the union of branches of two or more vessels
- Venous
 - Allows for alternative drainage of organs
- Arterial
 - Collateral circulation – alternative routes for blood to reach a tissue or organ



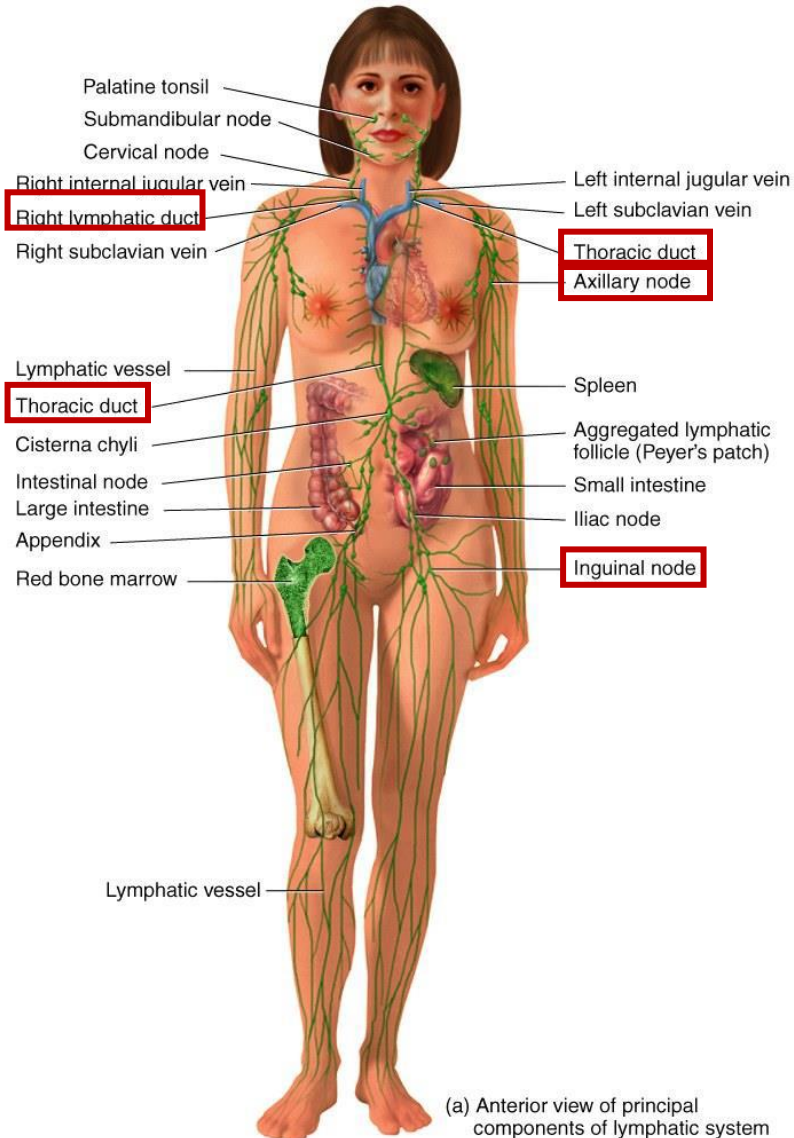
Lymphatic system

- Consists of lymphatic vessels, lymph nodes and lymphatic organs
- Drains excessive interstitial fluid
- Excessive interstitial fluid flows into lymphatic capillaries located amongst the tissues.
- Lymph eventually drains into lymphatic vessels.
- Located alongside lymphatic vessels are lymph nodes. Larger groups are found near the mammary glands, axilla and groin. Lymph nodes function as a filter for the fluid flowing through them.
- Eventually all lymphatic fluid drains back into the cardiovascular system via the right lymphatic duct and thoracic duct.
- Lymph nodes are usually removed during dissection and will most likely not be seen during the lab.



(b) Details of a lymphatic capillary

Lymphatic system



(b) Areas drained by right lymphatic and thoracic ducts

Area drained by right lymphatic duct

Area drained by thoracic duct